08 The Keystone Room

Predation meets competition

CORE PATH · BSC	EXTENSION · MSC	BEFORE THIS ROOM
about 35 min	**+ 30 min**	The Neighbours (05) and Predator (07) rooms, whose competitors and predator this room combines.
steps 1–4	steps 5–7	

OPEN THE ROOM

`https://evolinus.github.io/zooe/teaching/population-laboratories/` — then the Keystone tab

The Neighbours Room's two competitors and the Predator Room's predator in one community. Every run is simulated twice — as you set it and with the predator deleted — so a removal experiment is always on screen. This guide asks when a predator changes who survives, and why.

LEARNING OBJECTIVES

After the core path you should be able to

- Explain keystone predation: a predator that maintains coexistence by feeding on the dominant competitor.
- Use the predator-free counterfactual to measure a predator's effect on each species.
- Explain apparent competition, and why it can look exactly like resource competition.
- Show that a predator only acts as a keystone when it prefers the species that would otherwise win.

The extension adds

- Apply the three-species invasion criterion and check it against the room.
- Find how unequal the attack rates must be for keystone predation to work.
- Read the *Tolerance of predation* panel, r/a.

SUGGESTED ROUTES

- **40-minute practical (BSc):** steps 1–4 and questions 1–3.
- **Field-ecology seminar:** step 1 and 2 with Paine's *Pisaster* experiment (question 4).
- **MSc:** steps 5–7 and questions 4–6 (about 30 minutes).

How to work through it. Every step asks you to *predict* before you press anything, and to *explain* afterwards. Write the prediction down and commit to it: a wrong prediction you can account for teaches more than a right one you guessed. Runs are random, so where a step says so, run it more than once before you decide what happened.

1 Keystone predation

10 min

SET UP

Preset **Keystone predation**.

PREDICT FIRST

Without the predator, A would exclude B (as in the Neighbours Room). With a predator that attacks A four times as hard as B, what happens?

RUN

Run.

LOOK AT

the solid lines against their pale dashed predator-free twins, and the *Outcome* panel.

EXPLAIN

Which species does the predator help, and how?

2 Remove the keystone

8 min

SET UP

Preset **Remove the keystone**: identical parameters, predator absent.

PREDICT FIRST

What will happen to B?

RUN

Run.

EXPLAIN

What would a field ecologist conclude from this removal experiment, if they had not seen step 1?

3 Apparent competition

10 min

SET UP

Preset **Apparent competition**. Check that both competition coefficients really are 0.

PREDICT FIRST

The two prey do not compete at all. Can one still eliminate the other?

RUN

Run.

LOOK AT

What the predator is eating, and the predator's numbers.

EXPLAIN

How does B suffer from A's presence, through the predator? Why would field data make this look like competition?

4 Eating the wrong one

7 min

SET UP

Preset **Eating the wrong one**: the predator now prefers B, the weaker competitor.

PREDICT FIRST

Will the predator still maintain coexistence?

RUN

Run and compare with the dashed counterfactual.

EXPLAIN

Why is a predator not automatically a keystone? What must its preference be?

EXTENSION · MSC

5 The invasion criterion

12 min

SET UP

Preset **Keystone predation** (r = 0.5, K_A = 700, K_B = 400, α_{AB} = 0.5, α_{BA} = 1.5, a_A = 0.008, a_B = 0.002, e = 0.4, m = 0.6).

PREDICT FIRST

In a world of A and the predator, $N_A^* = m/(ea_A)$ and $P^* = (r_A/a_A)(1 - N_A/K_A)$. *Calculate both, then B's growth rate when rare:* $r_B(1 - \alpha_{BA}N_A/K_B) - a_B P^*$. Can B invade?

RUN

Compare with the *Invasion criteria* panel.

EXPLAIN

Explain in words why the predator holding A at 187.5 is what lets B in.

6 How lopsided must the plate be?

10 min

SET UP

Keystone predation; leave a_B = 0.002 and lower a_A step by step: 0.008, 0.007, 0.006, 0.005.

PREDICT FIRST

At what attack rate on A will B stop being able to invade?

RUN

Run each and watch the *Outcome* panel.

EXPLAIN

Why does the effect collapse well before the two attack rates are equal?

7 Tolerance of predation

8 min

SET UP

Keystone predation.

PREDICT FIRST

Each competitor can withstand at most r/a predators when rare. Calculate it for A and B.

RUN

Read the *Tolerance of predation* panel and the vertical marks.

EXPLAIN

How does the comparison of these tolerances with the predator densities each competitor supports decide the outcome?

DISCUSSION AND WORKSHEET QUESTIONS

1. BSc Paine removed the starfish *Pisaster* from rocky shores and diversity collapsed. Explain his result using this room.
2. BSc Two snail species show opposite trends in abundance. Give two explanations, one involving competition and one not, and suggest a field test to tell them apart.
3. BSc Why might removing a predator to “protect” a prey species backfire?
4. MSc Holt (1977) introduced apparent competition. Give a real example, for instance from biological control or invasive species.
5. MSc The predator here never switches to the commoner prey. How would prey switching change keystone effects?
6. MSc Explain how indirect effects make it hard to predict the consequences of removing one species from a food web.

CHECKING YOUR REASONING

Read this after you have written your own answers. These are what the model should show on most runs — not the only acceptable explanations.

- **Step 1.** All three persist; without the predator (dashed lines) A excludes B. The predator helps B by holding A down.
- **Step 2.** B disappears. A removal experiment reveals the predator's role only because it was compared with the intact community.
- **Step 3.** A eliminates B, with zero competition: A feeds the predator, and the predator, attacking B harder, drives B out.
- **Step 4.** No: preferring B, the predator hastens B's exclusion. A keystone must prefer the species that would otherwise win.
- **Step 5.** $N_A^* = 0.6/(0.4 \times 0.008) = 187.5$, $P^* \approx 45.8$, and B's growth when rare $\approx 0.148 - 0.092 \approx +0.06$: B can invade.
- **Step 6.** B's invasion rate turns negative between about 0.007 and 0.006, well before the attack rates are equal: the predator must press much harder on A.
- **Step 7.** A tolerates at most $0.5/0.008 \approx 62$ predators, B $0.5/0.002 = 250$. A on its own supports about 46 predators — close to its own limit, so A is held well below its carrying capacity, while B barely notices that many predators. That is what opens the door for B.